Young Jo(seph) Chung

PhD Student, Faculty of Information, University of Toronto

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Profile

PhD student researching agent-oriented requirements engineering for AI-enabled socio-technical systems under Prof. Eric Yu. Develops i* (iStar) Strategic Dependency and Strategic Rationale models to surface intentional actors, dependencies, vulnerabilities, and identity concerns when AI agents take on roles inside organizations, paired with empirical analysis of human–AI collaboration patterns at scale (29,585 pull requests across five AI coding tools). Two first-author papers accepted in 2026 (CAiSE; AIware).

Education

PhD in Information — Faculty of Information, University of Toronto 2025 – 2029

Master of Digital Experience Innovation — Stratford School of Business, University of Waterloo 2024 – 2025

Bachelor of Information Science — Faculty of Information, University of Toronto 2022 – 2024

Mathematical Statistics — Faculty of Mathematics and Statistics, York University 2018 – 2020

Publications

Chung, Y. J., Bae, K. E., & Yu, E. (2026). Modeling and Analyzing Identity-Sensitive Requirements in Human-AI Collaboration. In Proc. 38th International Conference on Advanced Information Systems Engineering (CAiSE 2026), Verona, Italy. Springer LNCS.

Chung, Y. J., & Hassan, S. (2026). Collaborator or Assistant? How AI Coding Agents Partition Work Across Pull Request Lifecycles. In Proc. 3rd ACM International Conference on AI-Powered Software (AIware 2026). ACM.

Honours & Awards

- SSC 2023 Case Study Award — Statistical Society of Canada & Bank of Canada (2023).

Research Experience

Research Fellow — Faculty of Information, University of Toronto Sep 2025 – Apr 2026; renewed May – Aug 2026

Supervisor: Prof. Eric Yu. Project: Agent-Oriented Modeling for Enterprise AI and Sustainability.

Student Research Trainee — Faculty of Information, University of Toronto May – Sep 2025

Supervisor: Prof. Eric Yu. Project: Agent-Oriented Conceptual Modeling.

- Developed i* (iStar) Strategic Dependency (SD) and Strategic Rationale (SR) models mapping the intentions, goals, and interdependencies of human and AI actors in enterprise settings.

- Constructed AS-IS models of existing processes (budget-traveller accommodation search, Excel-based accounting workflow, AI talent acquisition pipeline) to surface vulnerabilities and operational risks; designed TO-BE transformations integrating Agentic AI and Virtual Reality platforms.
- Conducted viability analyses and label propagation to simulate the impact of communication breakdowns and platform failures on high-level business goals and user satisfaction.
- Extended i* with identity-sensitive constructs (person-based, role-based, and relational identity dimensions) to capture requirements for transparency, authority preservation, and competence development under organizational AI adoption.
- Authored two first-author papers accepted to CAiSE 2026 (Verona) and AIware 2026.

Teaching

Teaching Assistant — RSM 8431 Social Network Analysis — Rotman School of Management, University of Toronto 2025 – 2026

- Graded MBA assignments on community detection, centrality, and network construction over the IWSPA-2022 phishing-email dataset and a 22-network Cisco Secure Workload corpus.
- Built a reproducible grading pipeline: ground-truth replication notebook, deterministic audit scripts, and a rubric-aligned grade-entry sheet to keep marks defensible and consistent across submissions.
- Held office hours and supported students on NetworkX, Gephi, and R/igraph, coaching interpretation of modularity, betweenness, and structural-hole measures.

Selected Pre-PhD Projects

HealthNet Ontario — Capstone Project — University of Toronto 2024

- Led a six-month Agile capstone analyzing health-data systems with emphasis on transparency and user-centered design; completed UX/UI for an Ontario Health application under industry mentorship.
- Built a document-grounded chatbot using OpenAI GPT-4 with a Pinecone vector index over custom-ingested documents.

Case Studies in Data Analysis Competition — Statistical Society of Canada 2023

- Led a three-month team analysis; performed mining, cleaning, and visualization in Tableau and R; produced a written report and poster presented at the Statistics Canada Annual Meeting seminar (SSC 2023 Case Study Award, see Honours).

Prior Professional Experience

Data Analyst, Maintenance and Control — Ontario Power Generation, Darlington Nuclear Generating Station 2016 – 2022

- Led audit-program planning and risk assessment across nuclear maintenance operations; elicited objectives and risk models from business-unit process owners — early exposure to stakeholder-driven requirements analysis in a safety-critical socio-technical setting.

- Performed data analysis to improve accuracy and accountability under high-stakes decision-making conditions; broke down strategic problems into insights and recommendations.

Market Research Analyst — Quantum & Baytree Real Estate Group, Toronto 2022 – 2023

- Defined business and IT requirements with stakeholders, conducted independent strategic analyses, and produced regular and ad-hoc reports and dashboards.

Instrumentation & Control, Maintenance and Control — Ontario Power Generation, Pickering Nuclear Generating Station 2013 – 2016

- Worked on design improvements, optimization, and automation of nuclear plant subsystems; ensured efficient process control with Allen-Bradley PLC and DCS systems; configured and designed Delta-V DCS software for extruder operation.

Certifications

- OISE Certification of Teaching English as a Foreign Language (TEFL).
- Microsoft Certified: Security, Compliance, and Identity Fundamentals.
- Certified Data, Library, Software Instructor — The Carpentries.
- Google Data Analytics Professional Certificate — Coursera.
- Virtual Engagement Program — Risk & Insurance Center and Royal Bank of Canada.

Skills

Research & Modeling: i* (iStar) / Tropos, agent-oriented conceptual modeling, requirements engineering, goal modeling, label propagation, qualitative analysis.

Data & ML: data visualization, data modeling, survey sampling, data mining, machine-learning pipelines (Naive Bayes, classification, evaluation), reproducible analysis notebooks.

Tools & Platforms: Python, R, SQL, JavaScript, Microsoft Azure, Tableau, NetworkX, OpenAI / Pinecone, Git.

Languages: English (native), Korean.